

SEMINDA FERNANDO

Software Engineer Undergraduate | Full Stack Developer

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PROFESSIONAL SUMMARY

Undergraduate Software Engineering student with hands-on experience in UI/UX design, backend development, and full-stack application development. Skilled in designing intuitive, user-friendly interfaces and building reliable, scalable backend systems through academic and personal projects involving both web and mobile applications. Proficient in working across the full software development lifecycle, from requirements analysis and interface design to server-side development, database integration, and deployment. Passionate about learning emerging technologies, solving complex problems, and delivering high-quality software solutions in collaborative team environments.

TECHNICAL SKILLS

Programming Languages: Java, Python, SQL, JavaScript (basic)

Frameworks & Libraries: Flutter, React (basic)

Databases: MySQL, MongoDB, Firebase (basic)

UI/UX & Design Tools: Figma, Adobe XD

Tools & Version Control: Git, GitHub, VS Code

Soft Skills: Leadership, communication, problem-solving, adaptability, critical thinking

KEY PROJECTS

Bus Go Mobile App — UI/UX Design

Design Tool: Figma

- Designed the user interface for a bus booking / tracking / scheduling mobile application.
- Created user flows, wireframes, and interactive prototypes using Figma.
- Focused on improving navigation, accessibility, and overall user experience.
- Analyzed existing transportation applications to identify usability gaps and opportunities for improvement.

Lost and Found Mobile App — Backend Development

Tech Stack: Node.js · Express · MongoDB · Backend Development

- Implemented CRUD operations for managing lost and found item records.
- Integrated MongoDB for efficient data storage and retrieval.
- Developed user authentication and authorization for secure access to application features.
- Structured backend architecture to ensure scalability, maintainability, and performance.
- Collaborated with the frontend team to integrate backend services with the mobile application.

Hostel Management System

Tech Stack: MongoDB · MySQL · Java · HTML · CSS · JavaScript

- Designed and developed a comprehensive hostel management system to automate student accommodation and administrative processes.
- Built responsive and user-friendly interfaces for administrators and users.
- Implemented backend functionality to manage student records, room assignments, and fee information.
- Applied software engineering principles and best practices to develop a maintainable, scalable, and organized codebase.
- Developed CRUD operations for managing students, rooms, and hostel records.

SmartCare Hospital — AI-Powered Predictive Healthcare System

Tech Stack: Python · Pandas · NumPy · Scikit-Learn · Matplotlib · Seaborn · SHAP/LIME · Academic Group Coursework (CCS3440 – Artificial Intelligence)

- Contributed to a 4-member team project analyzing a 1,000-record SmartCare Hospital dataset spanning patient demographics, clinical measurements, hospital operations, and billing data to build an AI-powered clinical/operational decision-support model.
- Owned Exploratory Data Analysis (Task 04): performed descriptive statistics, correlation analysis, and distribution analysis, and produced histograms, boxplots, scatterplots, and correlation heatmaps using Matplotlib and Seaborn to surface patterns in patient risk factors.
- Owned Machine Learning Model Development (Task 05): trained and compared multiple classification models (including Logistic Regression, Decision Tree, and Random Forest) using Scikit-Learn, tuning hyperparameters and benchmarking performance to support model selection.
- Collaborated with teammates on data preprocessing, model evaluation, and explainable AI (SHAP/LIME) to ensure the final prototype was interpretable for clinical decision support.

From Black-Box Predictions to Visual Intelligence: An XAI & 3D Visualization Framework for Vehicle Failure Management (Ongoing)

Tech Stack: Python · Flask · Three.js · Random Forest/XGBoost · SHAP/LIME · Final Year Project, SLTC Research University

- Co-developing a human-centered predictive maintenance framework for Toyota Aqua (Prius C) engine failure management, as part of a 4-member interdisciplinary Final Year Project team spanning Data Science and Software Engineering, supervised by Mr. Prabhath Samarasinghe and Dr. Chameera De Silva.
- Contributing to backend development: building a Flask-based system to integrate machine learning fault predictions (Random Forest, XGBoost), SHAP/LIME explainability outputs, and role-based alerts for drivers, technicians, and manufacturers, feeding processed data into an interactive Three.js 3D engine visualization module.
- Supporting the end-to-end system workflow — sensor data → ML prediction → XAI explanation → fault-to-component mapping → role-based alert → 3D visualization — for a project evaluated via System Usability Scale (SUS) scores and a ~30-participant user study on trust, understanding, and decision-making.

EDUCATION

- BSc (Hons) in Software Engineering — Sri Lanka Technology Campus, 2024–2027
- G.C.E. Advanced Level – Technology Stream, 2021
- G.C.E. Ordinary Level, 2018